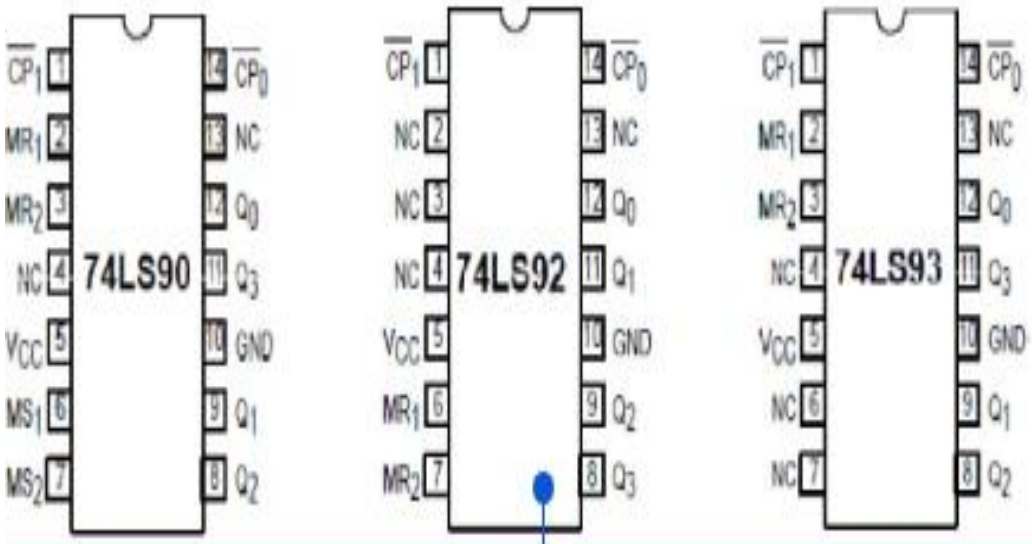
s compteurs **74LS90** (**modulo 10**), **74LS92** (**modulo 12**) et **74LS93** (**modulo 16**) sont des compteurs ynchrones composés de **4** bascules dont les connexions internes varient suivant le type du compteur.



$_0$	Clock (Active LOW going edge) Input to +2 Section
$_1$	Clock (Active LOW going edge) Input to +5 Section (LS90), +6 Section (LS92)
$_1$	Clock (Active LOW going edge) Input to +8 Section (LS93)
$_1$, MR_2	Master Reset (Clear) Inputs
$_1$, MS_2	Master Set (Preset-9, LS90) Inputs
)	Output from +2 Section (Notes b & c)
, Q_2, Q_3	Outputs from +5 (LS90), +6 (LS92), +8 (LS93) Sections (Note b)

LS90
MODE SELECTION

RESET/SET INPUTS				OUTPUTS			
MR_1	MR_2	MS_1	MS_2	Q_0	Q_1	Q_2	Q_3
H	H	L	X	L	L	L	L
H	H	X	L	L	L	L	L
X	X	H	H	H	L	L	H
L	X	L	X	Count			
X	L	X	L	Count			
L	X	X	L	Count			
X	L	L	X	Count			

H = HIGH Voltage Level
L = LOW Voltage Level
X = Don't Care

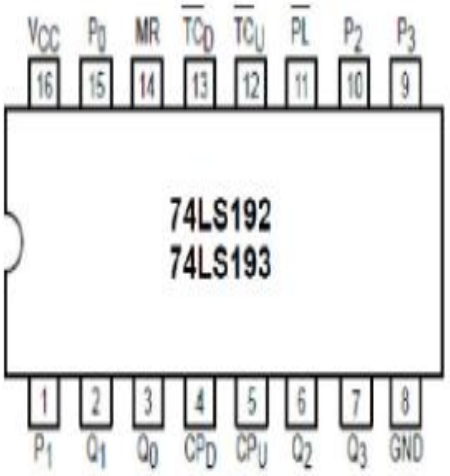
LS92 AND LS93
MODE SELECTION

RESET INPUTS		OUTPUTS			
R_1	MR_2	Q_0	Q_1	Q_2	Q_3

Remarque :

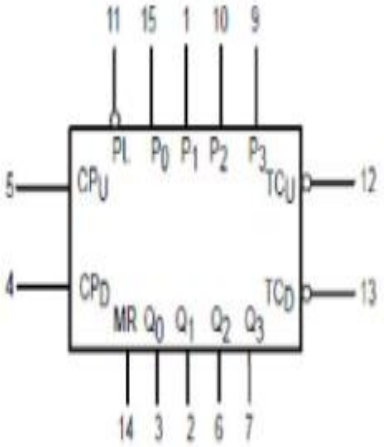
La 1ère bascule, diviseur par **2**, est séparée des **3** autres bascules qui sont interconnectées de telle sorte pour former un diviseur par **5**

CONNECTION DIAGRAM DIP



CP_U	Count Up Clock Pulse Input
CP_D	Count Down Clock Pulse Input
$\overline{MR}$	Asynchronous Master Reset (Clear) Input
$\overline{PL}$	Asynchronous Parallel Load (Active LOW) Input
P_n	Parallel Data Inputs
Q_n	Flip-Flop Outputs (Note b)
$\overline{TC}_D$	Terminal Count Down (Borrow) Output (Note b)
TC_U	Terminal Count Up (Carry) Output (Note b)

LOGIC SYMBOL



V_{CC} = PIN 16
GND = PIN 8

MODE SELECT TABLE

MR	$\overline{PL}$	CP_U	CP_D	MODE
H	X	X	X	Reset (Asyn.)
L	L	X	X	Preset (Asyn.)
L	H	H	H	No Change
L	H	⌐	H	Count Up
L	H	H	⌐	Count Down

L = LOW Voltage Level

H = HIGH Voltage Level

X = Don't Care

⌐ = LOW-to-HIGH Clock Transition